Number operation properties

1 State whether each expression is an example of the associative law, commutative law, reordering or distributive law.

a $15+5=5+15$

b $24\times 6=6\times 24$

c $54\div 9=9\div 54$

d $24+6+2=24+(6+2)$

e $126\div 3=(120+6)\div 3$

f $(15+6)\times 4=(15\times 4)+(6\times 4)$

Associative law

2 State whether the following statements are true or false.

a $4\times(3+5)=4\times 3+4\times 5$

b $4+(3\times 5)=4\times 3+4\times 5$

c $(4\times 3)+5=4\times 5+3\times 5$

d $(4+3)\times 5=4\times 5+3\times 5$

3 Fill in the boxes to complete the working out using the associative law.

a $35+29+11=35+(\square+\square)$

b $28+46-16=28+(\square-\square)$

4 Consider $27+48+13$.

a Which pair of numbers will be easiest to add together first?

b Fill in the boxes to complete the working out.
$27+48+13=27+(\square+\square)$

5 Fill in the boxes to complete the working out using the associative law.
$13\times 4\times 5=\square\times(4\times\square)$

Reordering

6 Evaluate $17+39+23$ using reordering.

7 Consider $52-24-12$.

a Is $52-12-24$ equal to the above expression?

b Which of the arithmetic rules can we apply to the expression to transform it into $52-12-24$?

8 Derek is evaluating the subtraction below. [redacted]oticed that it would be easier to subtract 27 from 47 first and rearranged the numbers using reordering.
Complete the following statement and then evaluate it:
$47-18-27=\square-27-\square$

9 Consider $56-18-26$.

a Which of the two numbers will be easier to subtract from 56 first?

b Complete the following statement and then evaluate it:
$56-18-26=56-\square-\square$

10 Evaluate $57-29-17$ using reordering.

11 Consider $48 \div 8 \div 2$.

a Is $48 \div 2 \div 8$ equal to the above expression?

b Which of the arithmetic rules can we apply to the expression to transform it into $48 \div 2 \div 8$?

Commutative Law

13 Consider $4 \times 13 \times 5$.

a Which pair of numbers will be easiest to multiply together first?

b Complete the following statement and then evaluate it:
$4 \times 13 \times 5=\square \times \square \times 13$

14 Evaluate $4 \times 17 \times 5$ using regrouping.

Distributive Law

15 Consider 35×23.

a Complete the following statement:
35×23 is the same as $35 \times(20+\square)$.

b Complete the following statement:
$35 \times(20+3)$ is the same as $35 \times \square+35 \times 3$.

c Which arithmetic rule explains the equality between 35×23 and $35 \times 20+35 \times 3$?

d Evaluate the expression.

16 Consider 26×29.

a Complete the following statement:
26×29 is the same as $26 \times (\square - 1)$.

b Complete the following statement:
$26 \times (30 - 1)$ is the same as $26 \times 30 - 26 \times \square$.

c Which arithmetic rule explains the equality between 26×29 and $26 \times 30 - 26 \times 1$?

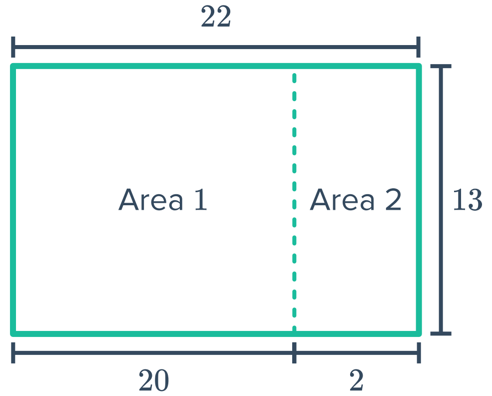
d Evaluate the expression.

17 Consider the following rectangle:

a Find the area of Area 1.

b Find the area of Area 2.

c Find the total area of the rectangle.

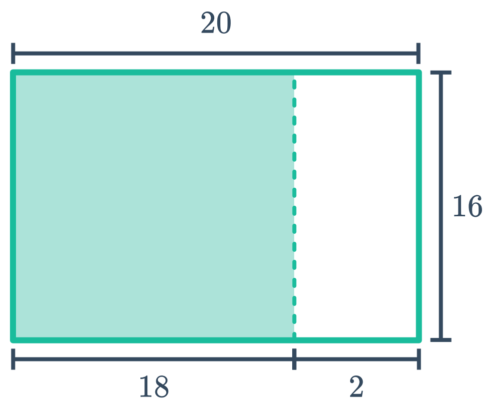


18 Consider the rectangle below:

a Find the total area of the rectangle.

b Find the non-shaded area of the rectangle.

c Find the shaded area of the rectangle.



19 Evaluate the following using the distributive law.

a 9×998 **b** 7×1002 **c** 11×124 **d** 6×412

20 Dave is evaluating $132 \div 11 \div 4$. He noticed that it would be easier to divide 132 by 4 first and rearranged the numbers using reordering.
Fill in the boxes to complete his working out.
$132 \div 11 \div 4 = 132 \div \square \div 11$

21 Consider $168 \div 7$.

a Fill in the box to complete the sentence.
$168 \div 7 = (140 + \square) \div 7$.

b Fill in the box to complete the sentence.
$(140 + 28) \div 7$ is the same as $140 \div 7 + \square \div 7$.

c Which arithmetic rule explains the equality between $168 \div 7$ and $140 \div 7 + 28 \div 7$?

d Evaluate the expression.

22 Consider $119 \div 7$.

a Fill in the box to complete the sentence.
$119 \div 7$ is the same as $(140 - \square) \div 7$.

b Fill in the box to complete the sentence.
$(140 - 21) \div 7$ is the same as $140 \div \square - 21 \div \square$.

c Which arithmetic rule explains the equality between $119 \div 7$ and $140 \div 7 - 21 \div 7$?

d Evaluate the expression.

23 Evaluate the following using the distributive law.

a $1616 \div 8$ **b** $396 \div 4$ **c** $522 \div 6$ **d** $585 \div 13$